

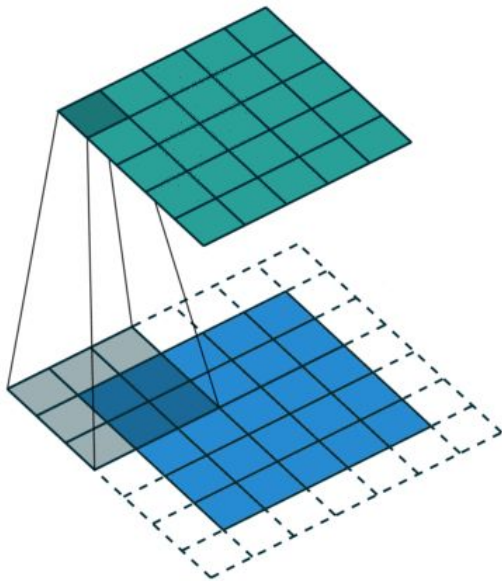
Multiscale and Hybrid Vision Architectures

Samsung Research Institute, Bangalore

8th June, 2026

CNNs:

Date



Build:

Convolution Layers, Activation, Pooling Layers, Fully Connected Layer.

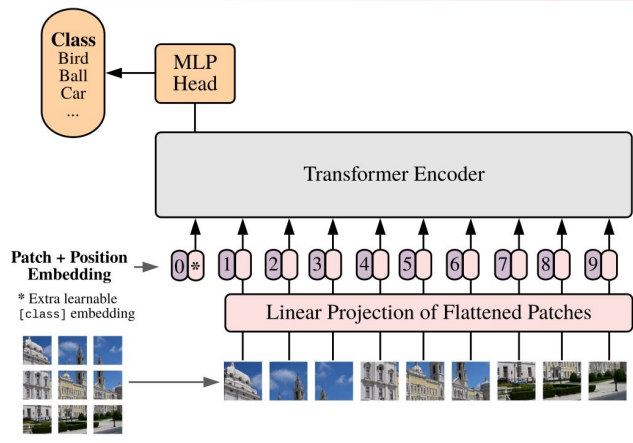
Properties:

Local Receptive Fields: Layers look at small, localized chunks of an image at a time rather than the whole thing at once.

Enforce strong inductive bias (translation invariance and locality)

Shared Weights: Instead of learning a new feature detector for every single corner of an image, a CNN learns one detector (like an "edge detector") and slides it across the entire image.

Spatial Hierarchy: The early layers detect simple things (edges, colors). The middle layers combine those edges into textures and motifs (eyes, wheels). The final layers piece those motifs into complex concepts (faces, cars).



Build:

Patch Embedding, Position Embedding, CLS token, Multi-Head Self Attention.

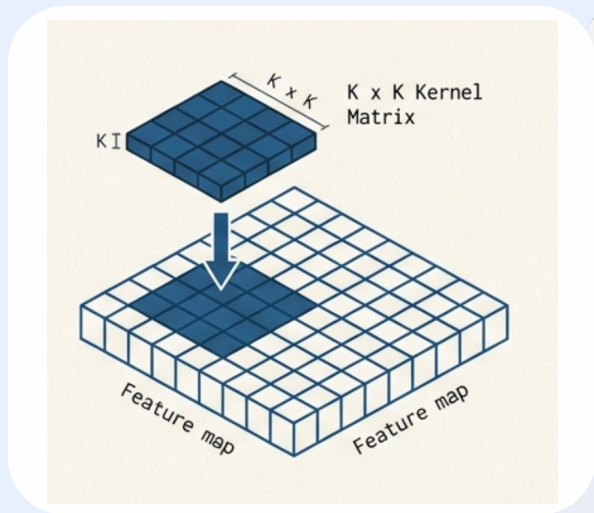
Properties:

A ViT views an image as a sequence of tokens, exactly like words in a sentence.

The model then uses Self-Attention to allow every token to look at every other token simultaneously.

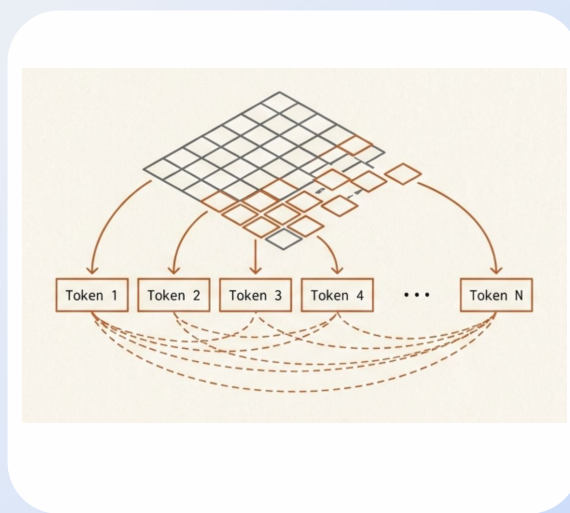
This allows the network to capture long-range dependencies and global context from the very first layer, something CNNs can only do after many downsampling steps.

The Core Paradigm Shift: Local Neighbourhood vs Global Sequences



CNNs:

- Receptive field: Localized, Grows with depth.
- Complexity Bounds: $O(H \times W \times C)$, Linear to resolution.
- Data dependency: Data efficient, works with small datasets.
- Scaling: Saturates early.
- Weakness: struggles with long range semantic dependencies.



ViTs:

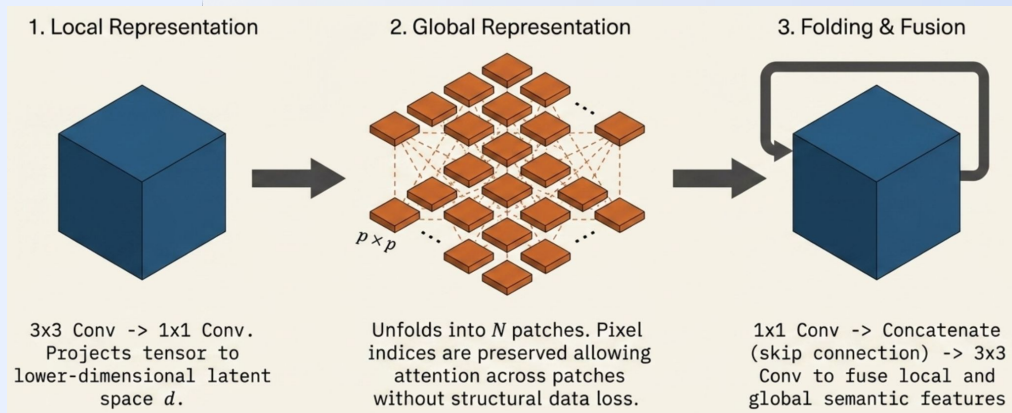
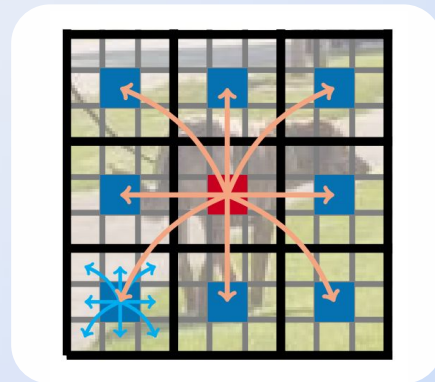
- Receptive field: Global from layer 1.
- Complexity Bounds: $O((H \times W)^2)$, Quadratic to token count.
- Data dependency: Data hungry.
- Scaling: Monotonic increase with compute.
- Weakness: Crushing memory/compute footprint for high resolution images.

Is it possible to combine the strengths of CNNs and ViTs to build a light-weight and low latency but highly capable network ?

MobileViT

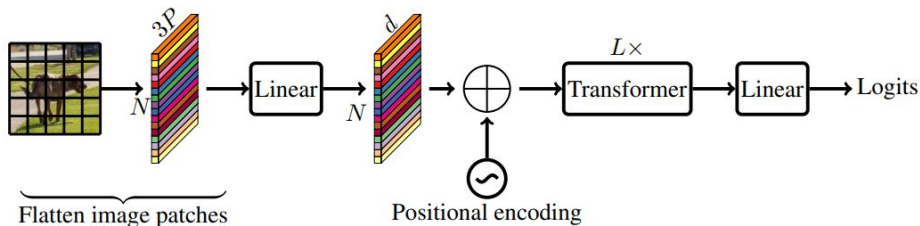
(Introduced by Apple AI Research)

- Light-weight, General Purpose, Low latency.
- Combines the benefits of CNNs (e.g., spatial inductive biases and less sensitivity to data augmentation) and ViTs (e.g., input-adaptive weighting and global processing).
- It treats the Transformer as a specialized layer within a standard convolutional network.
- MobileViT block replaces local processing in convolutions with global processing using transformers.

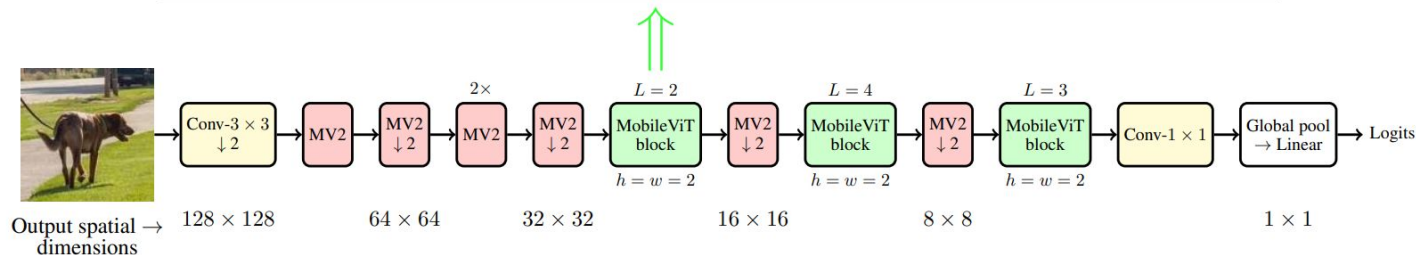
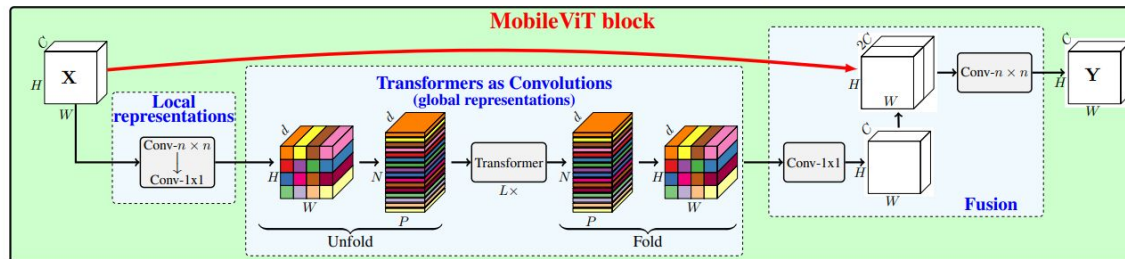


MobileViT: Architecture

Date



(a) Standard visual transformer (ViT)



(b) **MobileViT**. Here, $\text{Conv-}n \times n$ in the MobileViT block represents a standard $n \times n$ convolution and **MV2** refers to MobileNetv2 block. Blocks that perform down-sampling are marked with $\downarrow 2$.

Lets Trace the Tensor

Step 1: Local Convolution

input: $X \in \mathbb{R}^{H \times W \times C}$

$$X_L = \text{Conv}_{1 \times 1}(\text{Conv}_{n \times n}(X))$$

$$X_L \in \mathbb{R}^{H \times W \times d}$$

Step 2: Unfolding

$$X_U \in \mathbb{R}^{P \times N \times d}$$

$$N = \frac{HW}{P}$$

Patch size: $p \times p$

$$P = p^2$$

Step 3: Global Attention

$$X_G = \text{Transformer}(X_U)$$

$$X_G \in \mathbb{R}^{P \times N \times d}$$

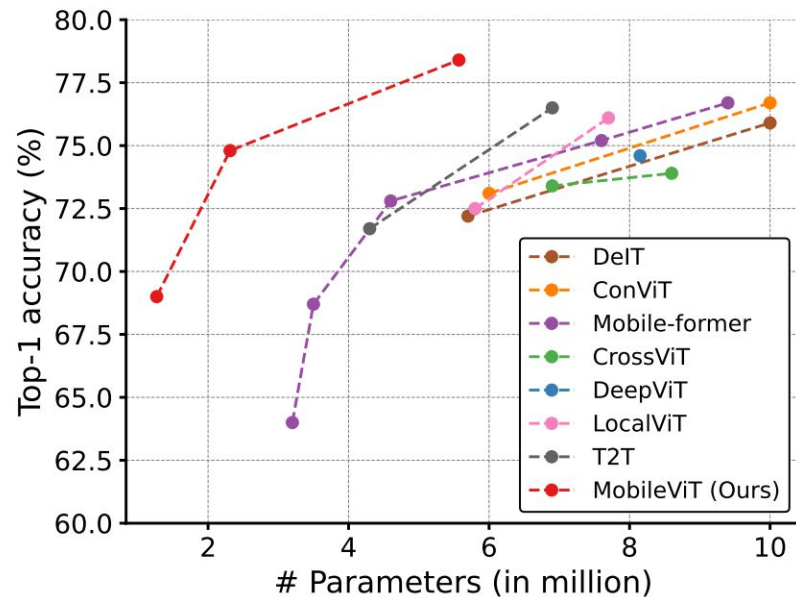
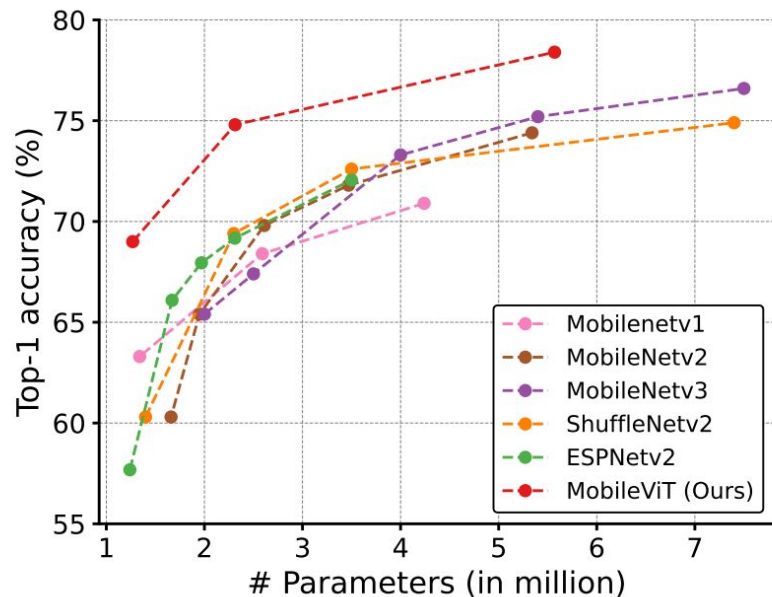
Step 3: Folding and Convolution Fusion

$$X_F = \text{Fold}(X_G) \quad \text{where} \quad X_F \in \mathbb{R}^{H \times W \times d}$$

$$Y = \text{Conv}_{n \times n}([X, \text{Conv}_{1 \times 1}(X_F)])$$

Comparison with CNNs and ViTs.

Date



Thank you



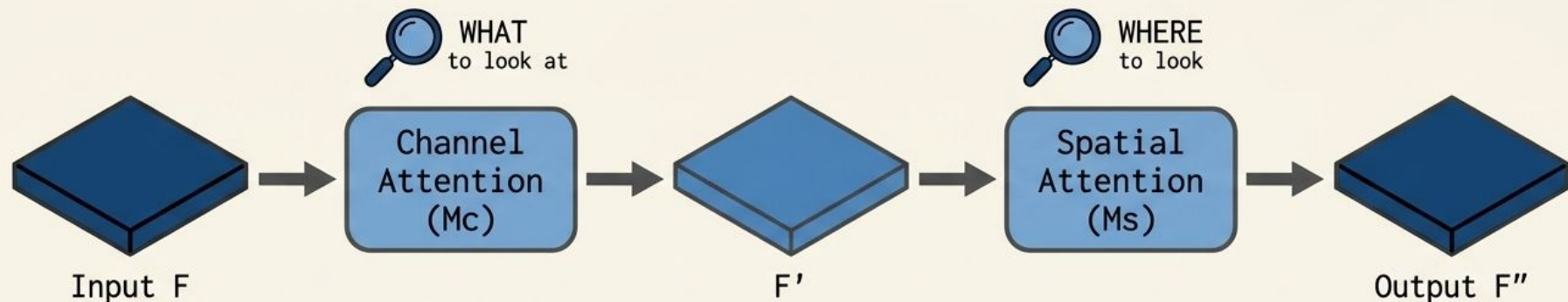
Dump:

Date

$$\mathcal{O}(P \cdot N^2 \cdot d)$$

$$\mathcal{O}(N^2 \cdot P^2 \cdot d)$$

CBAM: Plug-and-Play Sequential Attention



Channel Attention

$$Mc(F) = \text{sigmoid}(\text{MLP}(\text{AvgPool}(F)) + \text{MLP}(\text{MaxPool}(F)))$$

Identifies “what” is important. Aggregates spatial information via simultaneous Average and Max Pooling into a shared MLP.

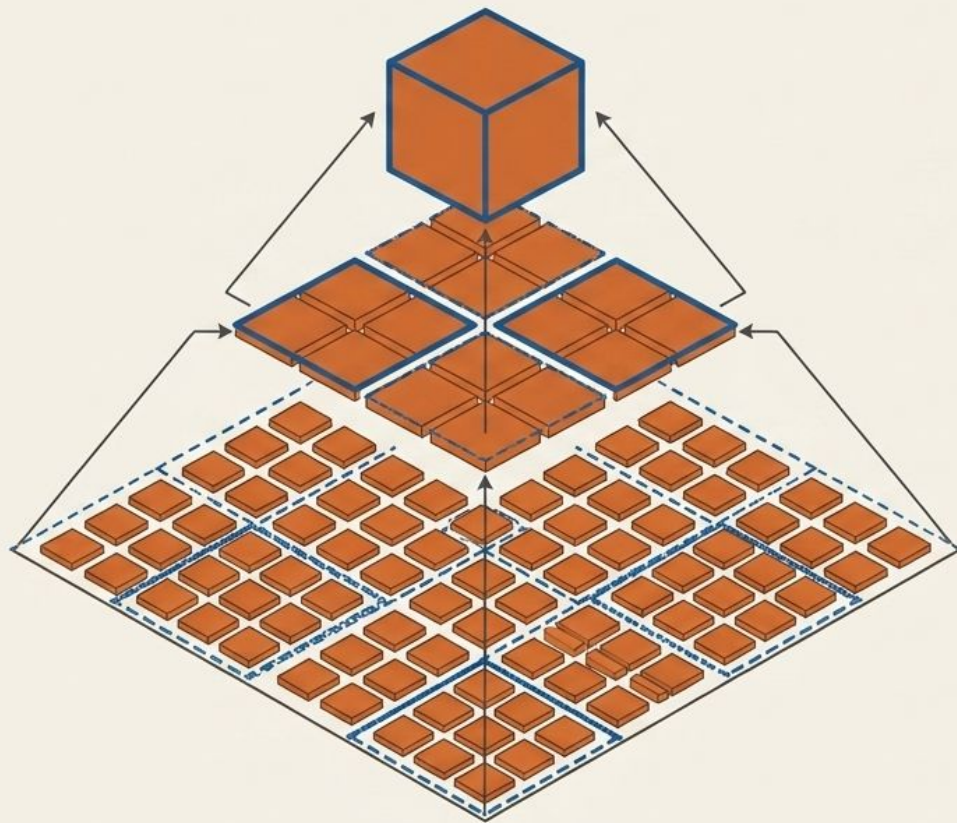
Spatial Attention

$$Ms(F') = \text{sigmoid}(f^{(7 \times 7)}([\text{AvgPool}(F'); \text{MaxPool}(F')]))$$

Identifies “where” to look. Applies a 7x7 convolution across concatenated channel-pooled maps.

Deployment: Drops directly into classic CNN backbones to boost multi-label parsing without retraining base parameters.

Scaling the Grid: Swin Transformer



The High-Resolution Bottleneck

Standard global attention fails on dense pixel prediction tasks (e.g., Mask R-CNN, volumetric CT scans).

Hierarchical Bounding

Swin restricts self-attention (W-MSA) exclusively to localized, non-overlapping windows of size $M \times M$ (typically $M=7$).

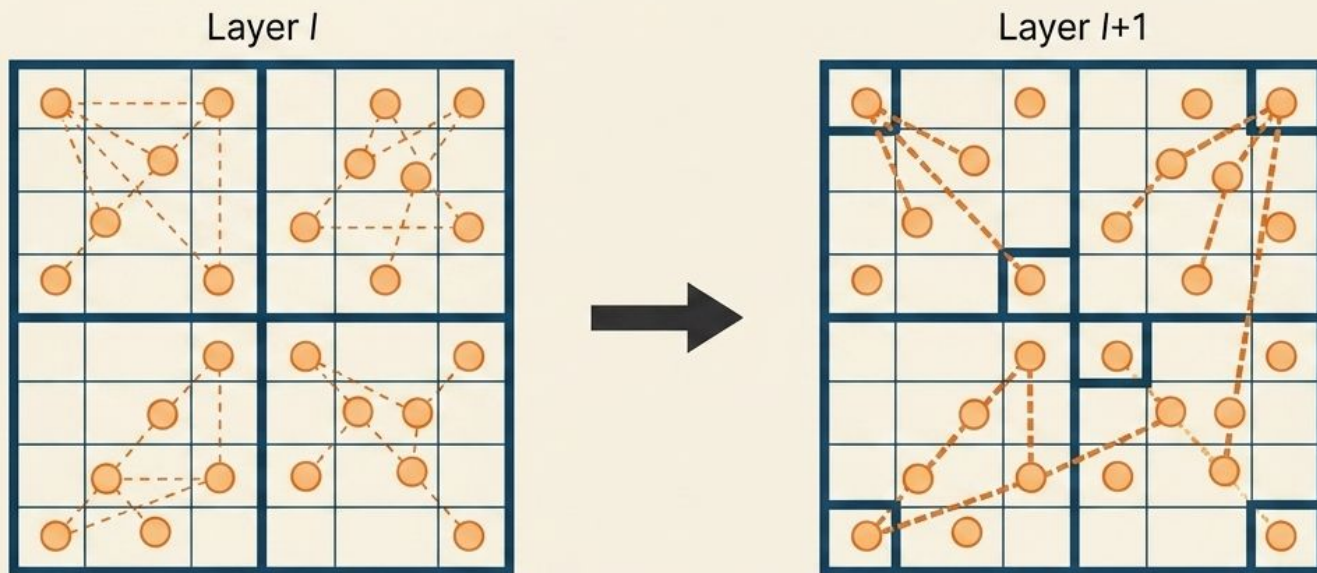
Computational Shift

By bounding attention, complexity drops from:

$$\begin{aligned}\Theta(\text{ViT}) &= 4HWC^2 + 2(HW)^2C \\ \Theta(\text{Swin}) &= 4HWC^2 + 2M^2HWC\end{aligned}$$

The second term becomes linear with respect to image area HW .

Breaking Boundaries: The Shifted Window Mechanism



The Problem

Strict windowing destroys the Transformer's primary advantage—global context.

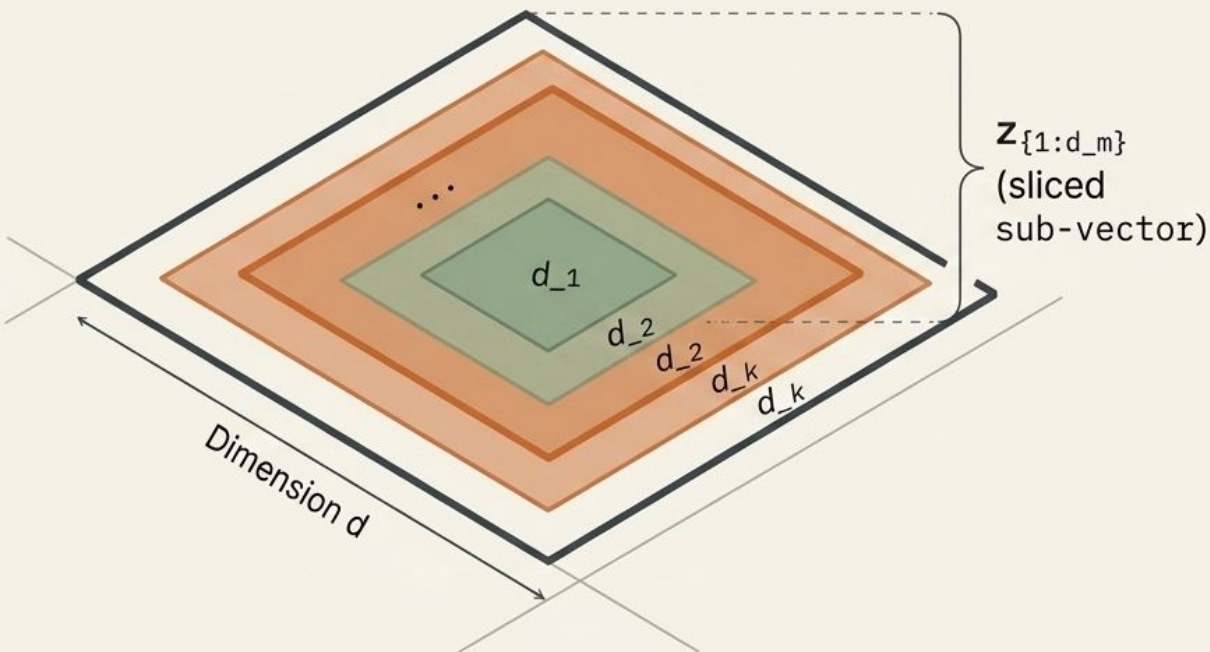
The SW-MSA Solution

Consecutive layers shift the window partition pattern down and right. This re-introduces global connectivity without the quadratic compute penalty.

Edge Handling

Fractional windows on boundaries are resolved via efficient cyclic shifting and masking to prevent non-contiguous regions from attending to one another.

The Rigidity Problem & Matryoshka Representation Learning (MRL)



The Constraint

Traditional models map inputs to static, fixed-dimensional embedding spaces, requiring complete retraining for different hardware deployments.

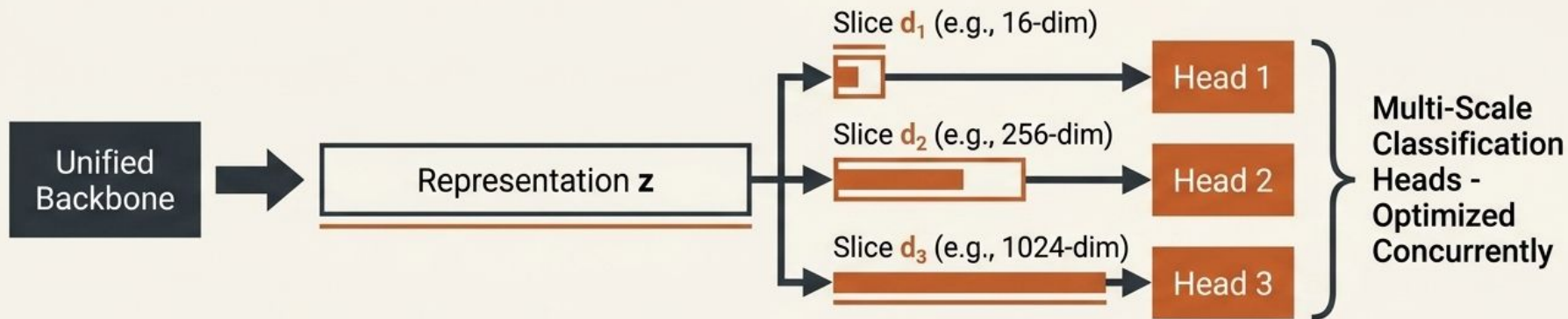
The Matryoshka Paradigm

MRL structures embedding vectors so early dimensions contain high-level semantic information, while later dimensions add finer granularities.

Adaptive Output

A single representation $\mathbf{z}$ can be sliced into sub-vectors. This allows one model to dynamically adapt embedding lengths based on immediate storage or compute limits.

Optimizing for Elasticity: The MRL Objective



The Loss Function

$$\mathcal{L}_{\text{MRL}}(x; \theta) = \sum_{m=1}^K c_m \cdot \mathcal{L}_{\text{ce}}(\mathbf{W}^{(m)} \cdot \mathbf{z}_{1:d_m}, y)$$

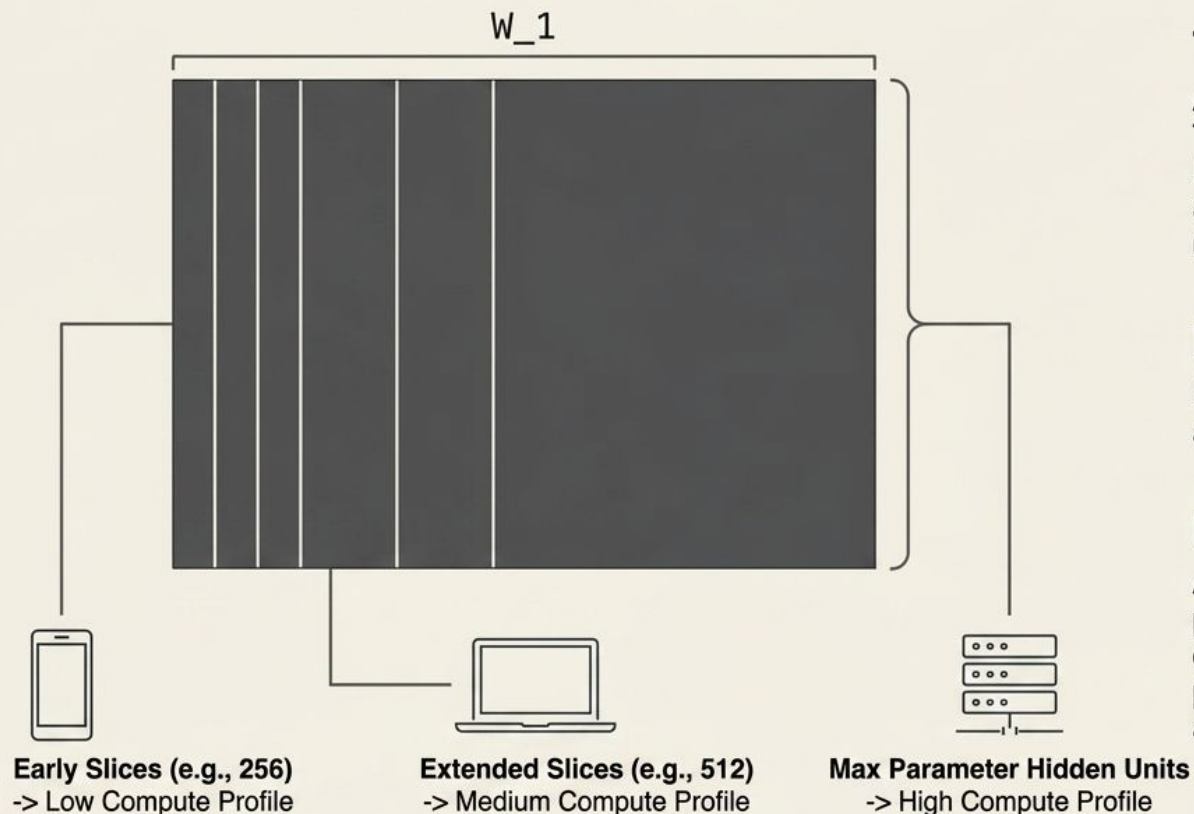
The network optimizes all chosen dimensional scales simultaneously.

Production Application

Adaptive Large-Scale Vector Search

Systems use the first few dimensions (**$\mathbf{d}=16$**) for ultra-fast candidate filtering, and full vectors (**$\mathbf{d}=1024$**) for precise final re-ranking.

MatFormer: Elastic Network Architectures



The Concept

Adapts MRL principles directly to Transformer internal weight structures. Instead of training separate ViT-Tiny, Small, and Base models, train a single universal structure.

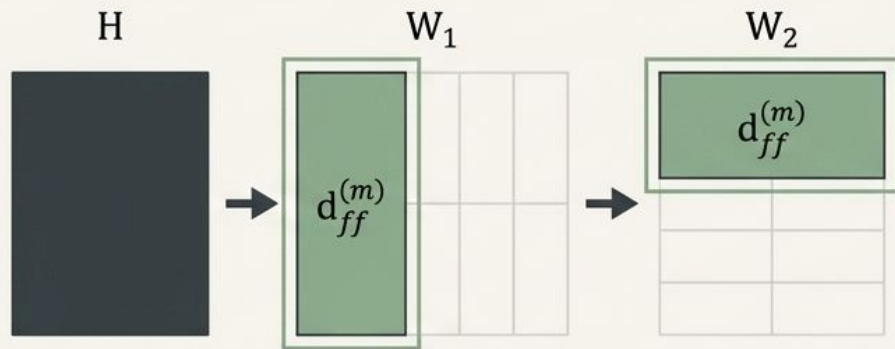
Nested Sub-Models

Hundreds of fully functional sub-models are nested inside the parent model.

Elastic Inference Extraction

Administrators can extract specific parameter sub-slices on the fly based on hardware constraints without performance distillation or architectural adjustments.

Executing the Elastic Extraction



Nested FFN Slicing

Given an intermediate dimension d_{ff} , smaller slices form independent, functional layers at runtime without retraining.

Mathematical Execution

$$\text{FFN}^{(m)}(H) = \text{GELU}(H * W_1[:, 1:d_{ff}^{(m)}]) * W_2[1:d_{ff}^{(m)}, :]$$

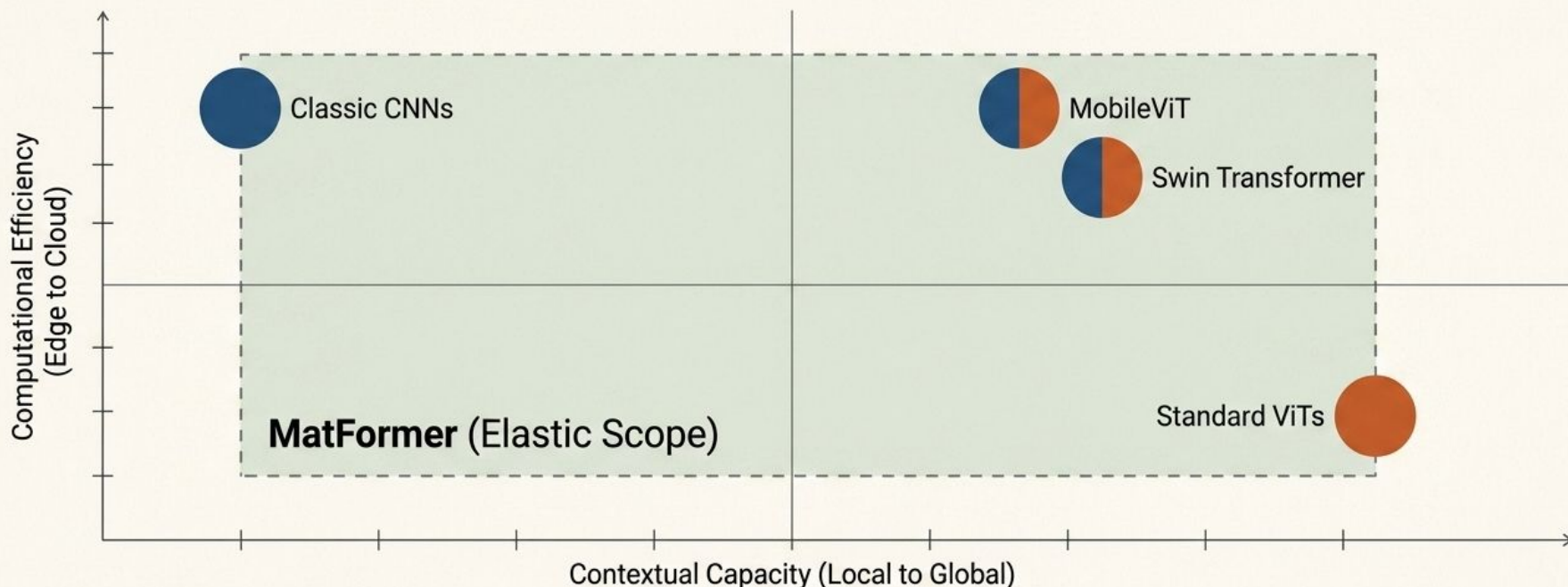
Nested Multi-Head Attention

Attention heads are ordered and regularized. Executing a subset (e.g., indices 0 to 4) yields a valid low-dimensional feature space.

Edge Engineering

Allows embedded applications to automatically downscale model width on the fly to conserve battery when energy drops below thresholds.

The Modern Architectural Landscape



Synthesis Insight

The evolution of vision networks has moved from selecting static trade-offs (CNN vs. ViT) to engineering structural synergy (MobileViT/Swin), ultimately arriving at true operational elasticity (MatFormer). Today's cutting-edge architectures don't just optimize compute—they dynamically adapt to it in real-time.